# Alternative-protein spending in India and Pakistan

## Conditional scenarios, exact methods and a separate elicitation supplement

Muhammad Umar Zafar

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# India and Pakistan: what spending scenarios can establish

**AI-assisted contribution draft, 22 October 2026. Human verification pending. Proposed for the unified report, subject to team agreement.**

India and Pakistan expose a basic difficulty in forecasting alternative-protein demand: the available numbers often describe something other than purchases. National food accounts measure supply, surveys ask about willingness, and company directories record activity. Each helps frame a question, but none supplies a national series of repeated purchases of meat, dairy or egg alternatives. This section therefore asks how spending might evolve under explicit assumptions, and which observations would make those assumptions more credible. It does not estimate animal-food displacement or returns on funding.

The starting point is GFI India and Deloitte's published estimate of India's 2022 smart-protein market: INR 351 crore across plant-based meat, dairy and eggs. Its forward domestic scenarios also include fermentation-derived meat. This is an external model estimate, not audited retail turnover. The source does not clearly state its price base or retail valuation. Our calculation assumes nominal 2022 rupees and treats the estimate as a retail-spending anchor; inflating it and converting at the 2025 exchange rate gives about $45.3 million. The retained evidence provides no equivalent Pakistan national sales anchor. That missing baseline must not be described as zero sales.[1]

The proxy model carries India's estimated spending per person forward with real income, urbanisation and an additional growth term. It transfers that path to Pakistan using relative income, urban share, animal-protein supply and industry-activity proxies. The relationships and parameter distributions are analyst choices, not coefficients fitted to alternative-protein purchases. The table reports the tenth, fiftieth and ninetieth percentiles across 10,000 assumption draws. These are conditional scenarios with untested calibration, in millions of US dollars at constant 2025 prices.[2]

| Country | Year | Lower | Middle | Upper |
|---|---|---|---|---|
| India | 2026 | 69 | 93 | 125 |
| India | 2030 | 102 | 188 | 334 |
| India | 2035 | 166 | 448 | 1,142 |
| Pakistan | 2026 | 1.5 | 3.6 | 6.6 |
| Pakistan | 2030 | 2.3 | 6.0 | 13.7 |
| Pakistan | 2035 | 3.6 | 11.8 | 36.1 |

The 2035 middle values correspond to roughly $0.28 per person in India and $0.040 in Pakistan. India's value is 1.22% of US plant-based retail spending per person observed in 2025; Pakistan's is about 0.17%. These are comparisons with a fixed historical benchmark, not shares of the US total market or comparisons with a 2035 US forecast. The US GFI/SPINS series covers broader categories, including prepared meals, so the ratio is illustrative rather than a matched product comparison.[3]

The results are sensitive to how the proxies are interpreted. The model's upper additional-growth assumption comes from growth in Indian job advertisements. Hiring activity may already reflect income growth; counting both could exaggerate expansion. A retained sensitivity that reduces this upper assumption lowers India's 2035 middle value from $448 million to $215 million. For Pakistan, removing the relative urban-share and animal-protein multipliers gives a middle value of $8.3 million and a tenth-to-ninetieth-percentile range of $2.6–25.3 million. Neither sensitivity is established as the correct specification. Together they show why the precision of the output exceeds the strength of the underlying relationships.[4]

Consumer research does not close that gap. Bryant and colleagues' online study found high stated willingness to buy plant-based meat in India, but its Indian sample skewed urban, affluent and educated. It measured hypothetical intentions, not subsequent purchases. The retained bilateral literature uses different samples and questions; it cannot support a national India–Pakistan ranking of willingness or conversion. Four outcomes need separate measurement: stated willingness, first purchase, repeat purchase after incentives end, and a measured reduction in animal-food purchases. A larger spending total alone cannot tell us which occurred.[5]

Policy and technical progress can change the conditions for those purchases, but their effects require separate evidence. India's 2024 BioE3 announcement includes smart proteins and proposed biomanufacturing and Bio-AI infrastructure. This establishes policy intent, not the delivery of a particular product or its consumer uptake. Pakistan has halal institutions; their existence does not establish food authorization for a particular novel product. Product-specific regulatory and religious questions therefore remain distinct. The present scenarios exclude cultivated meat; a future approval would require a newly specified scenario rather than an undocumented addition to this one.[6]

AI should be treated as a mechanism to investigate, not an automatic market-growth bonus. The retained research discusses laboratory optimisation and protein design, while company and policy material describe possible applications. These sources do not establish a country-level AI effect on alternative-protein spending. Improvements in conventional dairy or poultry could also change relative competitiveness. A useful test would compare validated output, production cost and retail prices against a credible non-AI baseline, while checking whether any effect is already counted in the model's other growth assumptions.[7]

The same discipline should guide talent recommendations. GFI India's talent study draws on 20 industry responses and 19 academic leaders or administrators; it offers specific leads, not a national workforce census. Job advertisements likewise do not measure successful hiring or skill shortages. A bounded next step is to test an employer-nominated technical constraint in India and assess the feasibility of an ingredient-application pilot in Pakistan, with facilities, staff and baseline performance checked first. These are proposals for learning, not a ranking of expected returns. Comparable sales and repeated-purchase evidence would do more to strengthen the demand argument than another layer of proxy complexity.[8]

# Methods appendix: India–Pakistan spending scenarios

**AI-assisted review draft, 22 October 2026. Human factual/code verification pending. Proposed joint-report methods contribution; definitions are proposed harmonization, not a recorded team agreement.**

## 1. Question, estimand and evidence classes

The calculation explores domestic alternative-protein spending in India and Pakistan in 2026, 2030 and 2035. Its intended scope is plant-based replacement foods and fermentation-derived human foods, excluding cultivated meat. The 2022 Indian anchor contains plant-based meat, dairy and eggs; the source's forward domestic categories include fermentation-derived meat. The model carries one aggregate forward and does not separately estimate entry of every proposed product category. Its interpretation must retain that limitation.

The proposed valuation is retail spending in market-exchange-rate US dollars at constant 2025 prices. **Retail valuation and the Indian source's nominal-2022 price basis are assumptions.** They are not explicit properties established by the original GFI/Deloitte summary. US coverage is broader and includes prepared meals. These differences prevent a fully matched cross-country market comparison.

Observed macro quantities, published external market estimates, analyst scenarios, model-elicited projections and missing evidence are recorded as different classes. The proxy scenarios estimate neither adoption probabilities nor physical protein volumes. FAOSTAT supply is an input proxy, not household consumption or alternative-protein demand. No model output is converted into an observed baseline.

The model was not fitted to a time series of alternative-protein purchases, and its future outcomes are unresolved. Calibration, predictive accuracy and probability coverage have not been demonstrated. Country-specific inference is limited to India and Pakistan.

## 2. Source selection and numerical inputs

The retained model reads **106 input rows** in `data/altprotein_proxy_inputs.csv`. Each row has an input ID, country/year, quantity, unit, source ID, tier, raw locator, capture reference/hash where available, and notes. The source collection is purposive; it was not a preregistered systematic review. Existing acquisition logs record unsuccessful and incomplete retrieval separately. Source dates and collection dates must not be conflated.

| Input family | Source/vintage and retained use | Main limitation |
|---|---|---|
| Indian market anchor and external scenarios | GFI India/Deloitte, 2022-context analysis, final-page domestic table: 351 crore in 2022; 2030 low/medium/high 7,475/12,779/18,963 crore | Published model estimates; price/valuation basis unstated; 2030 scenarios are not observations or probability bounds. |

| Input family | Source/vintage and retained use | Main limitation |
|---|---|---|
| US sales benchmark | GFI/SPINS 2025 plant-based retail sales, $7.9B; reported annual dollar change −2% | Broad US categories; nominal annual sales change is repurposed as a real non-income drift bound by assumption. |
| Population, GDP growth, inflation | IMF WEO October 2026: LP, NGDP_RPCH, PCPIPCH | Mixed historical estimates/outturns and projections; published horizon ends in 2031. |
| Urban share | WDI SP.URB.TOTL.IN.ZS, 2015–2025 | National urban definitions differ; future trend is extrapolated. |
| Exchange rate and income levels | WDI PA.NUS.FCRF; NY.GDP.PCAP.CN/CD | Market exchange rates, not PPP; observation years differ in the US–India elasticity construction. |
| Animal-protein supply | FAOSTAT 2023 protein supply, seven animal-food groups | Supply balance, not individual intake, willingness or replacement-product demand. |
| Industry activity | AltProtein.Jobs directory described as 2024 and annual vacancies 2023–2025 | Job-board coverage and country assignment are imperfect; directory records may overlap; not employment or skill shortages. |

The source ledger and source-specific rights should accompany extracted facts. Reproduction from the numeric input CSV does not require redistributing source workbooks, full PDFs, cached web pages or private team documents.

## 3. Exact implemented transformations

Notation: country `c` is India `I` or Pakistan `P`; year is `t`; `P_c(t)` denotes population **in millions**, `U_c(t)` urban share **in percent**, `I_c(t)` real GDP per person relative to 2022, `s_c(t)` annual dollars per person, and `M_c(t)` total spending **in million dollars**. Percentage units cancel in urban-share ratios.

### 3.1 Price conversion and anchors

Let `F = (1 + 5.4/100)(1 + 4.6/100)(1 + 2.0/100)`, using Indian consumer-price inflation in 2023–2025. Let `X = 87.158448` INR per USD, the retained rounded WDI 2025 period-average exchange rate.

```
Indian anchor A = 351 × 10 × F / X = 45.286640 million USD at 2025 prices
s_I(2022) = A / P_I(2022) = 0.031771 USD per person
s_US(2025) = 7.9 × 1,000 / P_US(2025) = 23.106847 USD per person
```

The factor 10 converts crore rupees to million rupees. The US 2025 dollars already use the selected price year. The Indian anchor is **a 2022 observation-year estimate converted to 2025 prices**, not a 2025 market observation.

### 3.2 Population and real-income paths

The retained IMF population series is used through 2031. Beyond it:

```
P_c(t) = P_c(2031) × [P_c(2031)/P_c(2030)]^(t−2031), for t > 2031
I_c(2022) = 1
I_c(t) = product over y=2023…t of [(1 + g_c(min(y,2031))/100)
                                    / (P_c(y)/P_c(y−1))]
```

Here `g` is real GDP growth. Holding the 2031 GDP and population growth rates through 2035 is an analyst extrapolation, not an additional IMF forecast. Retained 2035 income indices are 2.167826 for India and 1.254629 for Pakistan.

For each country, ordinary least squares on the eleven annual urban-share observations from 2015 through 2025 supplies a slope `b_c`. Future urban share is `U_c(2025) + b_c(t−2025)`. The function uses observed values through 2025. No estimated uncertainty or cross-country harmonization is added. Retained 2035 shares are 38.831347% and 43.289498%.

### 3.3 Parameter bounds

Four parameters are independently sampled from uniform distributions once per draw. The **same sampled parameters are reused across years and countries** within that draw; country/year results are therefore related, not independent samples.

| Parameter | Distribution/bounds | Construction and interpretation |
|---|---|---|
| Income elasticity `e` | Uniform(1, 1.810389…) | Lower bound assumes proportional income scaling. Upper bound attributes the entire US–India per-person spending gap to the market-dollar income gap. It is not an estimated causal elasticity. |
| Additional annual drift `d` | Uniform(−0.02, 0.17832570…) | Lower bound is US 2025 nominal retail dollar decline; upper bound is `(529/381)^(1/2)−1`, Indian vacancy CAGR for 2023–2025. Mapping these into future spending is an assumption. |
| Animal-supply weight `w` | Uniform(0,1) | Applies none to all of Pakistan's animal-supply ratio multiplicatively. |
| Pakistan industry factor `k` | Uniform(0.102303…,0.667385…) | Bounds are Pakistan/India directory-row density and vacancy density, both divided by 2025 population. |

The full-precision values are computed by the retained code from inputs, not from the rounded bounds above. Directory counts are 119 for India and 2 for Pakistan. 2025 vacancies are 529 and 58. Neither defines a complete national industry census.

For the upper elasticity bound, India's 2022 GDP per person in local currency is inflated by `F` and divided by `X`. US 2025 GDP per person is already in market dollars at 2025 prices. Then:

```
e_max = log[s_US(2025)/s_I(2022)]
        / log[GDPpc_US(2025)/(GDPpc_I(2022, INR) × F/X)]
```

The retained spending and income ratios are 727.301293 and 38.081810. They have a common price basis but different observation years. A same-price-basis conversion does not make them a matched contemporaneous comparison. Urbanisation, tastes, availability, category coverage and measurement also contribute to the observed gap; attributing it entirely to income is only a bounding assumption.

### 3.4 India and Pakistan equations

```
s_I(t) = s_I(2022) × I_I(t)^e × [U_I(t)/U_I(2022)] × (1+d)^(t−2022)

R_income(t) = [GDPpc_P(2025)/GDPpc_I(2025)]
              × [I_P(t)/I_P(2025)] / [I_I(t)/I_I(2025)]

s_P(t) = s_I(t) × R_income(t)^e × [U_P(t)/U_I(t)] × R_animal^w × k

M_c(t) = s_c(t) × P_c(t)
US comparison percentage_c(t) = 100 × s_c(t)/s_US(2025)
```

The 2025 Pakistan/India income ratio is 0.590535. `R_animal = 31.52/19.67 = 1.602440…`, from the sum of bovine meat, eggs, fish/seafood, milk excluding butter, mutton/goat, pigmeat and poultry protein supply. The industry and dietary multipliers are not independently identified effects. Their multiplication may double-count correlated conditions.

No Pakistan retail observation is used to fit or validate the transfer equation. Missing baseline evidence is never assigned a zero. No separate AI or policy multiplier is applied. No cultivated-meat entry term is present.

## 4. Draws, aggregation and output

The implementation uses Python's standard-library pseudorandom generator, `random.Random(20261015)`, and 10,000 draws. Every iteration draws `e`, `d`, `w`, `k` in that order. Results for each country/year are sorted. For quantile fraction `q`, compute `h=(n−1)q`, `i=floor(h)`, `j=ceil(h)`, and return `x[i]+(x[j]−x[i])(h−i)` with zero-based indexing. Output CSV values are formatted to six decimal places.

The main table reports q=.1, .5 and .9. They describe variation under chosen parameter distributions. They are **not** confidence bounds, a calibrated 80% predictive interval, or empirical household uncertainty. Display rounding must not imply greater accuracy. The chapter uses $448m and $11.8m for the retained 2035 medians 447.504341 and 11.759263.

## 5. Retained sensitivities and external benchmark

Sensitivities reset the same seed, retaining paired draws. Even a fixed or unused parameter is drawn before being replaced; this preserves pairing.

1. **Income/drift overlap:** annual real income growth from 2023 to 2025 is 6.548202%. The alternative upper drift is `d_max − income_growth × e_max = 5.977774%`; the lower bound stays −2%. India's 2035 p10/p50/p90 become 134.658168 / 215.060659 / 341.253412 million; Pakistan's become 2.382182 / 5.964900 / 11.690860 million. This is a sensitivity, not a fitted correction.
2. **Pakistan overlap:** set its relative urban share and animal-supply multipliers to 1 while retaining India's path, income ratio and industry factor. Pakistan's 2035 p10/p50/p90 become 2.575709 / 8.318219 / 25.341349 million.
3. **One parameter fixed:** hold each parameter at its lower and upper bound while drawing the others. India's 2035 median ranges are 332.438965–622.332639 million for elasticity and 127.357046–1398.084812 for drift. Pakistan's corresponding pairs are 12.293369 / 11.255477 for elasticity, 3.652403 / 40.094912 for drift, 9.273218 / 14.859778 for animal weight, and 3.468582 / 22.627744 for industry factor. These are pairs of conditional medians, not uncertainty intervals or variance decompositions. Higher elasticity need not increase Pakistan's transfer estimate because its relative income factor is below one.

The existing comparison converts GFI India's 2030 external scenarios using the **same 2022-to-2025 inflation and exchange-rate transformation** as the anchor, yielding 964.437694 / 1648.769136 / 2446.639731 million. This assumes the external scenarios share that source price base; the summary does not establish it. The fact that all retained proxy draws fall below GFI's low case is a conditional disagreement, not a validation test or proof of which forecast is correct. Preserve source rupee values alongside any converted comparison until the price basis is resolved.

## 6. Behaviour, policy and talent evidence

Behavioural evidence is read for the actual question and sampled population. Stated willingness, paid first purchase, unsubsidized repeat purchase and animal-food displacement are separate outcomes. The selective nine-study extraction package has no India finding and is not a systematic review of national adoption; the bilateral discussion draws on additional studies. Do not turn "not measured in these sources" into "does not occur."

Policy sources establish the dated instrument or announcement and its stated jurisdiction. Implementation, disbursement, product authorization and uptake need different evidence. Halal status and food authorization are distinct questions. This appendix does not provide a legal determination or assume a product's certification from general institutional powers.

Talent evidence separates job-board advertisements, directory entries, sampled employer testimony and national workforce measures. GFI India's study's 20 industry responses and 19 academic participants are study cohorts. Its findings may motivate a pilot; they do not identify a nationally optimal training portfolio. Recommendations in the chapter are author proposals that require baseline and outcome checks.

## 7. Validation, reproducibility and remaining holds

Run `python scripts/reproduce.py` from the package root. The packaged engine `scripts/build_altprotein_proxy.py` reads the 106 aggregate input rows and writes `build/proxy_scenarios.csv`. Its private acquisition/transcription mode has been removed; the

mathematical engine, random draw order and output format are preserved. No source acquisition or model call is required.

The 22 October package verification regenerated all 127 rows byte for byte against `data/reference_proxy_scenarios.csv`. Independent denominator and sensitivity checks are included in `tests/test_reproduction.py`. Clean-folder verification is recorded in the release receipt. Reproduction establishes consistency with the specified assumptions, not empirical validity.

Human reviewers must resolve the Indian price/retail basis, judge the crosswalk and proxy assumptions, verify included sources/data/code, and sign off the final claims. A new national sales series should be documented as new evidence and evaluated explicitly rather than silently replacing the original anchor. The separate supply benchmark's out-of-sample performance does not validate alternative-protein spending. Full source/capture pointers and rights limits are in claim map and source ledger.

# Measurement contract and proposed harmonization

**Proposed for review, 22 October 2026. No team agreement is asserted.**

| Strand | Outcome / channel | Technology and category | Geography / time | Uncertainty meaning |
|---|---|---|---|---|
| Regional proxy | Intended domestic retail spending; the Indian source's retail basis is assumed | Plant-based replacement foods plus fermentation-derived human foods; no cultivated meat; aggregate anchor limits category detail | India/Pakistan; 2022 Indian estimate converted to 2025 prices; 2026/2030/2035 scenarios | Percentiles under analyst-selected parameter distributions; no demonstrated coverage |
| Regional panel | Headline retail spending; excludes foodservice and conventional staples | Plant-based replacements, all dairy alternatives and other included plant-based replacements; headline excludes fermentation/cultivated technologies | India/Pakistan; 2026/2030/2035; million market-exchange-rate USD at 2025 prices | Percentiles of ten accepted response medians by requested model; not predictive bounds |
| Separate panel US arm | Original all-category plant-based retail question; additional technology questions have their own scope | Broader US plant-based basket, with distinct cultivated and fermentation questions | US methodological comparator in the author's supplement | Same descriptive run-median aggregation; prompts and supplied evidence differ by region |
| Historical US benchmark | GFI/SPINS observed retail sales | Broad basket including prepared meals | Fixed 2025 observation; per-person denominator | Source estimate; not a 2035 US forecast |
| Conventional food evidence | Supply balance or reported household food quantities | Animal foods and conventional categories | Source-specific year, sample and geography | Measurement/sampling properties; does not establish alternative-protein purchases |

For any other contribution to the unified report, record the actual executed question, product basket, channel, currency, price year, source vintage and uncertainty measure before comparing it. A common label introduced after collection does not make different protocols equivalent.

The main research question is what the evidence supports about demand and what would reduce decision-relevant uncertainty. Keep stated willingness, paid first purchase, repeat purchase and animal-food displacement separate. Model output, policy intent, company activity and missing data are different evidence classes. Do not sum regional results into a global market, pool proxy draws with model responses, or attribute disagreement to one cause without a design that identifies it.

# Separate forecast-elicitation supplement

**AI-assisted methods draft, 22 October 2026. Human verification pending. Authored for Muhammad Umar Zafar. This is a separate protocol and sample, not an accepted joint result.**

## 1. Purpose and limits

The exercise records repeated model-generated projections under specified prompts and execution conditions. It can describe within-workflow repetition and between-workflow disagreement. It cannot establish forecast calibration, independent expert opinion, causal effects of AI on food demand, or why a provider's outputs differ from another's.

The approved target was ten accepted responses per requested model and region: two models across India, Pakistan and the US, giving six cells and 60 accepted responses. All registered cells reached ten accepted responses. "Complete" describes that target, not scientific acceptance.

## 2. Instrument and country scope

The US questionnaire preserves the author-designated source text received for this execution. The source DOCX and derived text hashes are retained. Byte agreement proves provenance of this supplied version; it does not prove that every run in another study used exactly that prompt, background material or software. The original instrument and regional adaptations are local private inputs because public redistribution rights are not established.

| Arm | Outcome and product scope | Conditioning |
|---|---|---|
| India/Pakistan | Retail plant-based meat/seafood replacements; all dairy alternatives; other in-scope plant-based replacement products. Headline excludes fermentation and cultivated technologies, conventional staples and foodservice. | Country-adapted questions plus frozen public evidence summaries. |
| US | Original broad all-category plant-based retail question. Separate cultivated and fermentation questions retain their original retail/foodservice and pet-food inclusions. | Original questionnaire; no added project market figures in the US evidence pack. |

Forecast years are 2026, 2030 and 2035. Headline unit is **million USD at constant 2025 prices**. Regional category medians must sum to headline medians; marginal tail quantiles are not required to sum. All dairy alternatives in the questionnaire are broader than the milk-only calibration used in a separate household simulator. Results must not be relabelled as milk-only or as that simulator's validated inputs.

The regional instrument also requests stated-intent conversion, Bass coefficients, price-parity conditions, AI-attributed manufacturing-cost decline and separately defined trade/listing/external-estimate proxies. Those variables have different units and resolution conditions; none is a substitute for measured headline retail demand. The India/Pakistan national headline is presently unresolvable from a complete observed series. A null parity year denotes the stated unknown/never condition, not zero. AI manufacturing-cost decline is not retail pass-through. A “first legal sale” question is distinct from a sales total.

## 3. Evidence selection and possible anchoring

Country evidence packs were frozen on 19 October 2026. India includes the external GFI/Deloitte estimate and forward scenarios, conventional household consumption context, sample-specific stated-intent research, trade-proxy limitations and policy context. Pakistan includes conventional household context, an upstream trade observation, the same non-Pakistan intent comparison, and explicitly identified India-only external estimates. Neither pack includes project household-model results, internal price ratios, project market forecasts or unit-record data.

The packets themselves disclose that some cited sources rely on earlier catalogue review rather than a fresh freeze-day retrieval. An empty Comtrade capture means missing data, not zero trade. Upstream protein concentrates and listing counts are not alternative-protein retail sales. The GFI forward scenarios may anchor respondents; their broader technology scope differs from the regional headline.

The US evidence file does not receive the added regional facts. Therefore “all runs had the same evidence pack” is false across regions. Similar isolation rules also do not establish equal executed retrieval. Evidence-conditioned elicitation differs from a blank-slate protocol that omits supplied historical observations and external forecasts. This exercise does not isolate that effect and is not a controlled experiment on evidence provision.

## 4. Execution, model identity and dates

Requested model labels were `gpt-6-astra` and `claude-opus-5-5`. Every accepted Claude receipt reports `claude-opus-5-5`. Accepted Astra receipts have an empty `reported_models` field; their identity is documented as requested, **not independently confirmed by provider-returned model metadata**. Different service names do not establish independent architectures or independent evidence.

| Requested model | Accepted-run time window, UTC | Accepted cells |
|---|---|---|
| gpt-6-astra | 19 October 2026, 18:40:58–19:50:24 | Ten India, ten Pakistan, ten US |
| claude-opus-5-5 | 20 October 2026, 19:15:23 to 21 October 2026, 15:12:55 | Ten India, ten Pakistan, ten US |

Each attempt uses a fresh temporary working directory, no resumed conversation and no prior model answers. Claude is configured with WebSearch/WebFetch, restricted mode, an explicit system instruction, no user/project settings and an empty strict MCP configuration. Codex is configured to disable shell, project instructions, memories, apps, skill discovery and delegation, and to use read-only sandboxing with

provider-event auditing. The runner documents that Codex has no demonstrated blanket end-user tool allowlist; observed prohibited command/file/MCP/delegation use is rejected. These are implementation and configuration claims, not proof that every possible contamination route was absent.

The retained receipt audit establishes configured web access for completed recorded attempts. It does **not** establish that all runs executed the same searches, fetched the same pages or had equivalent tool behavior. Search parity requires a separate event-level extraction and is not asserted here. Prompts, provider events and answers stay private; their hashes and sanitized metadata can support integrity checks without redistributing their contents.

## 5. Attempts, exclusions and protocol change

The receipt inventory contains **79 recorded attempts**, not 79 demonstrated completed model calls:

| Requested model / region | Accepted | Rejected | Execution error | Incomplete RUNNING record |
|---|---|---|---|---|
| Astra / India | 10 | 0 | 0 | 0 |
| Astra / Pakistan | 10 | 0 | 0 | 0 |
| Astra / US | 10 | 0 | 0 | 0 |
| Claude / India | 10 | 7 | 1 | 0 |
| Claude / Pakistan | 10 | 0 | 1 | 0 |
| Claude / US | 10 | 8 | 1 | 1 |
| **Total** | **60** | **15** | **3** | **1** |

Seven rejected Claude India receipts cite a first-sale-year validation rule. Six Claude US receipts cite an unapproved reported model, and two cite JSON parsing. The three execution errors are retained separately. The RUNNING record lacks a command receipt; its existence alone establishes neither a sent call nor completion. No attempt is retroactively renamed accepted to make the design look cleaner. `data/panel-attempts.csv` contains sanitized record states, model identities, dates, reason categories and receipt/parsed hashes.

On 20 October, a parser amendment allowed a cultivated first-sale year accompanying a “no” answer to be normalized to null. The original supplied year is retained as `cultivated_conditional_year_dropped`. Before that amendment, seven of eleven Claude India attempts had been rejected for this condition alone, compared with zero of the thirty Astra attempts. Previously rejected receipts remained rejected. Ten accepted responses per cell therefore do not imply equal selection mechanisms. The amendment, failures and retries may affect the accepted distributions; this supplement does not estimate that selection effect.

The runner validates units, finite values, year coverage, monotone quantiles, category coherence and prohibited source/tool contamination. It retains failures and uses fresh retries. Configuration defaults include bounded validation attempts and a 900-second timeout; actual slot histories can exceed one default batch through continuation. The attempt table, not a default-limit description, is the executed count record. No rerun was performed for this drafting task.

## 6. Aggregation and retained results

For each model, region, measure and year, the summary takes each accepted response's stated P50 and computes empirical percentiles across those run medians. It does **not** pool all response P10/P50/P90 values into one predictive distribution. Original quantiles are retained in the numeric response archive. Missing conditional values are excluded for that measure and `n_nonmissing` is reported; missingness is not filled with zero.

Sort the run medians as `x[0]…x[n−1]`. For fraction `q`, let `h=(n−1)q`, `i=floor(h)`, `j=min(i+1,n−1)`; return `x[i]+(x[j]−x[i])(h−i)`. For ten nonmissing runs, the median is the mean of the fifth and sixth ordered values, P10 is `0.1×first + 0.9×second`, and P90 is `0.9×ninth + 0.1×tenth`.

For a purely illustrative ordered set of ten run medians `1,2,…,10`, this rule gives P10=1.9, P50=5.5 and P90=9.1. These example numbers explain the interpolation; they are not study results or a model's own predictive quantiles.

| Region | Requested model | 2035 P10 of run medians | Median of run medians | P90 of run medians |
|---|---|---|---|---|
| India | Astra | 324 | 370 | 422 |
| India | Claude | 84.2 | 157.5 | 190.7 |
| Pakistan | Astra | 27.6 | 37 | 60 |
| Pakistan | Claude | 7.96 | 9.35 | 15.32 |
| US | Astra | 10,500 | 10,650 | 11,000 |
| US | Claude | 6,880 | 7,000 | 7,320 |

All values are million USD at 2025 prices; each cell has ten accepted, nonmissing headline medians. These are descriptive workflow outputs, **not calibrated predictive intervals**. No provider weights or grand pooled median are introduced. The India/Pakistan panel excludes fermentation whereas the proxy model includes it; their differences cannot be attributed solely to model optimism or evidence conditioning. The US arm remains in this separately authored methods supplement and does not expand Umar's main regional chapter into a US research claim.

## 7. Relationship to other forecasting studies

Any comparison requires the other study's executed-method record and author approval. Product scope, prompts, supplied evidence, execution date, tool use, model identity, number of submissions and aggregation can all differ. An envelope formed from minimum P10 and maximum P90 is not the same statistic as percentiles of repeated-run medians. An internally aggregated submission must not be counted as several equivalent observed runs.

A terminology decision after execution cannot change those differences. Discuss compatible summaries side by side with their definitions; do not pool them into a larger supposedly independent panel. This package contains no collaborator draft, private result table or claim of another author's approval.

## 8. Reproduction, rights and acceptance

Run `python scripts/reproduce.py` from the package root to reconstruct the **headline table** from `data/panel_runs.json`. The packaged `scripts/build_panel.py` implements the linear quantile rule above; it checks all six ten-run cells and rejects invalid units, duplicate slots and non-finite, negative or crossed quantiles. The reproduced 18 summaries are compared with `data/reference_panel_aggregates.json` at absolute tolerance 1e-12. This route uses only the Python standard library and no private answers, questionnaire text, provider logs, credentials or new model calls.

The complete elicitation is not independently rerunnable from a public package while questionnaire rights remain unresolved. An integrity hash does not grant redistribution rights. Retain original records privately and state this reproducibility boundary. Publicly proposed rows and metadata still require a content/rights review and human verification.

No other author's acceptance or agreement is claimed. Human factual/code review, author approval and source-specific release review are outstanding. Joint inclusion requires a separate editorial decision. Source pointers and hashes are recorded in the companion claim map and provenance files.

# Limitations and AI-use statements

**Draft, 22 October 2026. Human verification pending.**

## Main-text limitations

The India–Pakistan results are conditional spending scenarios. The Indian anchor is an external modeled estimate with an unstated price and valuation basis; the analysis assumes nominal 2022 rupees and retail value. Pakistan lacks a comparable verified national baseline in the retained evidence. Proxy coefficients and distributions are analyst choices, with possible overlap among income, urbanisation, animal-protein supply and industry activity. The reported simulation percentiles have no demonstrated probability coverage. Macro projections beyond 2031 extend the last available growth rates. The US comparison uses a different product basket and a 2025 observation rather than a matched forecast-year benchmark. National supply, stated intention, paid purchase, repeat purchase and animal-food displacement remain different outcomes. The source collection is selective and does not establish that evidence absent from this collection is absent everywhere.

The separate model-elicitation supplement is a repetition and disagreement exercise under recorded workflows. Its regional prompts, evidence conditioning, execution dates, tools and technology scope can differ from other contributors' studies. Model versions and attempted-run selection also matter. The accepted set excludes rejected and errored attempts, and a parser rule changed during collection. The spread of run medians is not a predictive interval. Similar answers do not prove calibration or independent evidence; different answers do not isolate the effect of a model, prompt or AI assumption. The supplement is authored as Umar's work and is not represented as an accepted joint result.

## Current, truthful AI-use statement

AI tools assisted source discovery and summarization, code development, quantitative scenario construction, structured forecast elicitation, and drafting/editing. The separate elicitation requested `gpt-6-astra` and `claude-opus-5-5`; provider identity evidence is reported in its methods rather than assumed. Codex assisted this 22 October writing and methodological audit, including inspection of retained calculations and execution receipts. The human author's scope choices and approvals are recorded separately from factual verification. Human verification of the included claims, citations, statistics, quotations, datasets and code is pending. These materials are review drafts and must not be represented as final or accepted programme outputs.

This statement describes known workflow categories. Before final submission, the human author must complete a tool/version/task register for substantive earlier AI use as well; this drafting pass cannot certify the full historical record.

## Final statement template — do not use until completed

> AI assistance was used for [verified list of tools and substantive tasks]. [Named humans] checked [scope and dated review records], including the cited originals, numerical transformations and executable code. Remaining limitations are reported in [locations]. [Insert the programme's required attestation only after it is true.]

The programme policy requires human verification of **all** included AI-produced or AI-edited claims, citations, statistics, quotations, datasets and code. Prioritize load-bearing claims, but do not narrow the requirement to those claims alone. Code review includes reading, running and testing. Agent checks do not substitute for human review. Scholarly peer review may remain pending and labelled after factual/code verification; that label is not an alternative to verification.

Private review drafts may retain explicitly marked unverified material for review. Final submission/public release must omit it or complete the verification. Source: Sentient Futures Project Incubator AI Use Policy, 20 August 2026, Verification, Transparency and Attestation sections. The policy document is not bundled; participants should consult their programme copy.

## Human review register fields

Claim ID; exact manuscript sentence/table; original-source locator; source vintage; numerical transformation/code revision; reviewer name; review date; disposition (verified/corrected/omit/unresolved); restrictions; author sign-off. A blank reviewer field remains pending. No reviewer identity or approval is inferred from possession of a file.

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1. GFI India and Deloitte, *Smart Protein Economic Analysis*, 2022-context analysis, methodology and final-page domestic-market table, original report. Precise publication date and price base are unstated; the upload path is not a publication date. Source mapping: C01–C03 in claim map.↩
2. Retained `altprotein_proxy_inputs.csv`, `altprotein_proxy_scenarios.csv` and `build_altprotein_proxy.py`; exact equations and assumptions in proxy methods. IMF WEO October 2026 and World Bank WDI provide macro inputs. Mapping: C04–C06.↩
3. GFI US retail market data, 2025 sales; IMF population; retained comparison rows. The US anchor is $7.9 billion, or $23.106847 per person. Mapping: C07–C08.↩
4. Retained 2035 sensitivity rows `market_value_drift_overlap_adjusted` and `market_value_without_urban_animal`; see methods for the exact changes. Mapping: C09–C10.↩
5. Christopher Bryant et al., "A Survey of Consumer Perceptions of Plant-Based and Clean Meat in the USA, India, and China," *Frontiers in Sustainable Food Systems* 3 (2019), doi:10.3389/fsufs.2019.00011; retained bilateral review and measurement audit. Mapping: C11–C12. The selective nine-study extraction package is not presented as a systematic national review.↩
6. Government of India, BioE3 Cabinet approval, 24 August 2024; Pakistan, *Pakistan Halal Authority Act, 2016*, especially establishment and functions. These are dated institutional sources, not a legal determination for a particular product. Mapping: C13–C14.↩
7. Dated source review in the retained AI-mechanism review, including Cosenza et al., doi:10.1002/elsc.202300005, and Watson et al., doi:10.1038/s41586-023-06415-8. The former's primary full text was not freshly recovered for this drafting pass. Mapping: C15.↩
8. Amrutha Girivasan and Chandana Tekkatte, GFI India, *A comprehensive analysis of India's talent landscape for the smart protein sector*, physical PDF page 10. The official website listing is dated 19 December 2025; the PDF itself is undated. Retained AltProtein.Jobs extracts and the author's capacity proposals are mapped in C16–C17. Pakistan's proposed pilot is a recommendation, not an observed programme.↩